



STUDENT INNOVATIVE PROJECT 2025-2026

COMPLETION REPORT



PROJECT TITLE:

**AI-Powered IoT Wearable for Stress and Sleep Management
for the Mental Well-being of Students and Professionals**

SIP ID: 2526S0124

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BONAFIDE CERTIFICATE

Certified that this Project Report titled **"AI-Powered IoT Wearable for Stress and Sleep Management for the Mental Well-being of Students and Professionals"** was submitted by **Mr. S. Abdullah, Mr. A. Alston Samuel Peter, Mr. A. Mohammed Arshad**, who carried out the work under our supervision. Certified that to the best of our knowledge the work reported herein follows all the guidelines prescribed by the University during and after the implementation of the project.

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We acknowledge with thanks the eight student volunteers from Anna University who participated in our wearable data collection sessions. Their contributions of recordings across a comprehensive range of physiological states including normal, stress, sleep, stress combined with sleep, post-workout, and dehydrated were fundamental for training our TinyML model.

Finally, we express our recognition of the EdgImpulse platform, the open-source Arduino and Zephyr RTOS communities, and the researchers whose published datasets (WESAD, PhysioNet) provided valuable insights into physiological signal processing.

ABSTRACT

Psychological issues such as chronic stress and disrupted sleep have become more common among student and working communities. While there are commercially available wearables that track some of the basic wellness indicators, very few, if any, offer comprehensive stress assessment, along with sleep tracking and real-time edge inference. The objective of this project was to address this recognized problem by developing an IoT-enabled multi-sensor wearable device that would classify a person's physiological condition in real time using a TinyML model running on the device itself.

The hardware platform uses the Arduino UNO Q and is made up of two modules, which include an Arduino processing module that utilizes a microcontroller and a Linux operating system module. Five different types of sensors have been used in the design of the hardware system, which include MAX30102 used for PPG and measuring heart rate and SpO₂ levels, MPU6050 for inertial measurements involving accelerometer and gyroscope readings, AD8232 ECG front-end, Grove GSR sensor for galvanic skin response, and finally EMC1101 temperature sensor. Feedback from the device is available via the use of SSD1306 OLED screen. A PCB for the system was designed online.

Data from real-time physiological sensors for seven different states were acquired from humans by using the prototype, and a TinyML classifier using a neural network was trained via the EdgeImpulse tool. The quantized int8 model that was produced was then tested and deployed directly on the hardware.

The Senti web application is a complementary application that serves as the interface for the wellness tracker. It retrieves the physiology classification from the API, shows live sensor values and real-time electrocardiogram waveform, and utilizes the agentic artificial intelligence (in particular, GPT-4o) for offering customized, interactive wellness tips. These wellness tips are tailored depending on the physiological state of the user. The entire system helps achieve UN Sustainable Development Goal 3 (Good Health and Well-Being).

Keywords: *IoT wearable, TinyML, edge computing, stress detection, sleep monitoring, ECG, PPG, GSR, EdgeImpulse, mental health, Arduino UNO Q, agentic AI.*

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INTRODUCTION

Modern society is increasingly faced with issues related to mental health. Students have to deal with various pressures in the form of academic workload, competitiveness in career opportunities, and lack of financial security, whereas professionals face long working hours, strict performance criteria, and inadequate rest. Chronic stress may cause disorganization of the sleep cycle, impaired cognition, a compromised immune system, and even result in heart disease in some cases [1]. According to a major study conducted by WHO, anxiety and depression result in economic losses to the world economy to the tune of one trillion dollars per year, which highlights the importance of identifying stress in society [4].

Existing wearables used by consumers such as smart watches and activity trackers tend to focus on tracking the number of steps taken, energy burned, and basic heart rate readings. The few that incorporate both electrodermal (GSR) measurements, electrocardiograms (ECG), and photoplethysmography (PPG) in one low-power wearable device are rare [5]. Even fewer perform on-device machine learning inferencing where sensor readings are processed on the device itself; instead, they send raw data to the cloud to be analyzed, a process that may suffer delays and security concerns due to internet connectivity [6]. Importantly, personalized conversational AI insights triggered by physiological feedback are lacking.

The project supported by Centre for Sponsored Research and Consultancy (CSRC), Anna University, within the scope of Student Innovative Project, 2025-2026 (SIP ID: 2526S0124), aims to overcome these shortcomings by designing an inexpensive multi-sensor based body worn system that performs real time classification of physiologic state using on-board machine learning [2, 7].

1.1 Motivation

The first reason for undertaking this research is the observation that among the student population of Anna University, during examinations there seems to be an increase in the incidence of health center visits and symptoms being attributed to stress, like headache, inability to sleep, heart palpitations, and feeling anxious. A wearable device that could detect any physiological precursors to these clinical signs before they manifest themselves and provide ways to combat them may be a worthwhile preventive measure.

The second reason for pursuing this project relates to the issue of improving health technology access. Smartwatches that give a rough approximation of one's level of stress tend to be priced anywhere between INR 15,000 to 80,000, besides depending upon proprietary cloud-based algorithms for its functionality. In comparison to this, our prototype that is entirely made up of off-the-shelf products has a sensor cost of less than INR 2,000, which amounts to nearly a magnitude of two reductions.

1.2 Scope of the Project

The scope of this project includes various domains:

- (a) Purchase of hardware, circuit design, and assembling a functional prototype;
- (b) Developing firmware for acquiring sensor data through the Bridge/RouterBridge approach on the Arduino UNO Q;
- (c) Collecting an original dataset from eight subjects
- (d) Training, testing, and deploying a TinyML model using the EdgeImpulse software tool;
- (e) Designing a complementary web application known as Senti, including a live dashboard and an AI-driven chatbot.

Explicitly, the scope does not include certification as a medical device, extensive clinical validation, and developing an app to be used in practice. These are reserved for future work.

LITERATURE REVIEW

2.1 Physiological Stress and Sleep Monitoring

Physiological responses in the face of stress trigger the HPA axis, which results in the release of cortisol and an increase in sympathetic nervous system arousal. Some of the physiological measurements that have been found to correlate positively with stress levels include increased heart rate (HR), decreased heart rate variability (HRV), increased Galvanic Skin Response (GSR), also referred to as Electrodermal Activity (EDA), as well as changes in electrocardiogram (ECG) morphology. On the other hand, sleep is marked by increasing HRV, stable GSR readings, as well as unique ECG signals during both non-REM and REM phases. By using multiple physiological parameters from a single device, we can employ multivariate classification, which tends to outperform univariate approaches.

2.2 Existing Wearable Systems

Rostami (2024) demonstrated stress detection in real-time using LSTM models with PPG signals on the Raspberry Pi with accuracy of 93.4%. Nevertheless, the limitation of using a single sensor and a complete processor with Linux operating system capabilities was a bottleneck in the implementation of such devices in wearables. Srivastava et al. (2024) proposed the EdgeML approach on microcontrollers for the detection of stress and sleep through HRV analysis, proving the effectiveness of edge inference in devices that possess resource limitations. According to a literature review conducted by Al-Atawi et al. (2023) of 34 papers regarding stress monitoring systems involving machine learning, Internet of Things, and wearable sensors, sensor fusion resulted in an 8-15 percentage point performance improvement.

In Abd-alrazaq et al. (2024), a systematic review and meta-analysis were conducted on wearable AI systems for the purpose of stress recognition among students, and the researchers reported a pooled sensitivity of 82.3% and specificity of 84.1%, which implies a substantial room for improvement, especially in regard to generalization. In their review on stress recognition and monitoring approaches, Pinget et al. (2024) found that the most discriminative stress indicator is GSR, while HRV and skin temperature indicators play a significant role as well. According to Abellán et al. (2019), there is justification for an edge computing architecture as it lowers the round trip latency from 200 to 800 ms (cloud computing) to less than 5 ms (edge computing).

2.3 TinyML and On-Device Inference

With the emergence of TinyML technology, which allows for machine learning computations to be carried out using microcontrollers with only kilobits of RAM, it has become possible to conduct neural network inference off the cloud. Dedicated software such as TensorFlow Lite Micro and EdgeImpulse provide methods of quantization (INT8/FLOAT32) and model pruning, which have been shown to reduce models' sizes 4-8x without significantly impacting their performance.

The Arduino UNO Q microcontroller based on STM32H7 and capable of running hardware floating point calculations is especially suitable for conducting inference in INT8 format due to its support by the EdgeImpulse Arduino library.

2.4 Agentic AI for Wellness

Advances made recently in large language models (LLMs) have made possible agentic AI systems that can reason about contexts through sensor inputs, make personalized recommendations, and conduct multi-turn conversations. Models such as GPT-4o and Claude Sonnet have demonstrated excellent performance in mental health support cases where the context is given in a structured form using physiological information. By implementing this sophisticated approach in the Senti application, the physiological state and important sensory information currently being monitored are incorporated into the system prompt for the LLM, enabling contextually appropriate micro-interventions to be performed.

2.5 Identified Research Gap

Based on the comprehensive literature review, no existing system concurrently achieves the following:

- (i) Five-modality physiological sensing (PPG, ECG, GSR, IMU, temperature) on a single, low-cost microcontroller-class device;
- (ii) Seven-class physiological state classification, which includes less-explored states such as Stress-Sleep co-occurrence and Dehydration;
- (iii) INT8 quantized on-device inference with a peak RAM consumption of merely 44 bytes;
- (iv) The integration of an agentic large language model through a complementary web application. This project is specifically to address and bridge this identified research gap.

METHODOLOGY

The project employed a methodology that followed an iteration consisting of ten phases covering hardware purchasing, firmware development, data collection, model training, and software implementation.

3.1 Hardware Procurement and Circuit Design

Selection of components was based on considerations such as low cost, compatibility with the Arduino UNO Q through I2C/SPI ports, and physiological relevancy. The entire set of sensors was purchased from ZBOTIC, Robu.in (Macfos Ltd.), AJANTHA Electronics, and Modi Electronics within the duration of January 2026 to March 2026, at an overall cost of INR 23,999.40 from nine bills, as shown in the budget utilization table. Custom printed circuit boards (PCBs) were created with online PCB platforms, incorporating the sensor connectors, a 7805 linear dropout (LDO) regulator and the Arduino UNO Q onto one board.

3.2 Firmware Development

This system implements the concept of dual-processing based on the Bridge and RouterBridge communication scheme that comes with the Arduino UNO Q board. It utilizes the C++ code for implementing the process of continuously reading all five sensor values in real-time using a frequency of 50 Hz. The read-out data is then delivered through the USB serial bridge to the Python process (main.py) that runs in Linux environment. This process uses the Python's thread-safe deque, creates batches of 50 samples each and stores it in the CSV file named as sensor_data.csv along with the timestamps of ISO 8601 format.

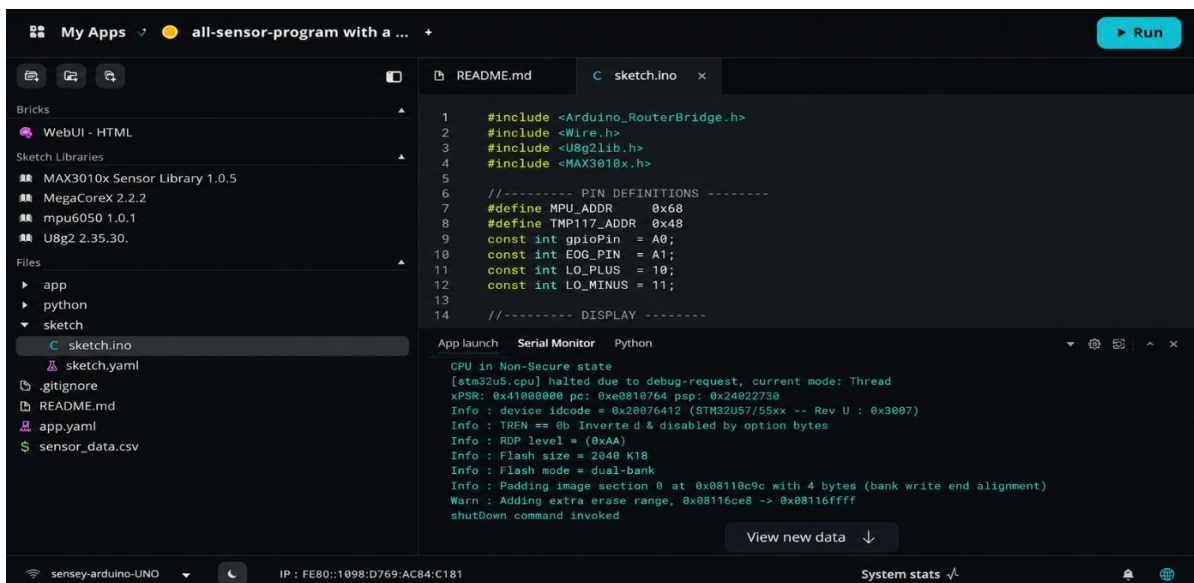


Figure 1: Arduino App Lab Interface (For UNO Q)

3.3 Sensor Data Acquisition Pipeline

The following data is outputted per sample using the CSV format: timestamp, PPG red channel, PPG IR channel, ax, ay, az (acceleration readings in g), gx, gy, gz (gyro readings in deg/s), GSR raw ADC readings, AD8232 ECG output, leads_off Boolean indicator, and temperature (°C). The GSR was designed to include a 500 ms delay for settling purposes due to noise that could occur in the system because of capacitive coupling in the analog power rail. In addition, the SSD1306 OLED screen had to implement an I2C driver for its functionality since the Adafruit GFX library did not work well with the Zephyr HAL implementation in the STM32 platform.

3.4 Data Collection Protocol

The experiment involved eight individuals who are all students of Anna University and aged 19 to 23 years old. Every experimental session was conducted in the laboratory located on the third floor of the IST Department. The prototype was worn on the non-dominant hand, while the placement of the electrodes was done according to Lead I setup. The physiological states were elicited in the following way, apart from them an additional 12 individuals (6 college students and 6 working professionals), data was generated using generative AI:

Normal: 10 minutes of seated rest following a 5-minute adaptation period.

Stress: Stroop color-word interference task and arithmetic task under time pressure for 10 minutes.

Sleep: Light dozing / relaxed supine posture in a dark room for 15 minutes.

Stress-Sleep: Cognitive load task immediately followed by supine rest, capturing the stress-to-sleep transition.

Post-Workout: 5 minutes of stair-climbing followed by 5 minutes of recording.

Dehydrated: Recording after participants had abstained from water for 4 hours.

Disconnected: Deliberate sensor detachment to capture artefact baseline.

Data quality was checked continually in real time using the OLED screen and the Python logging window. Sessions that presented significant artifacts due to high levels of motion or poor connection

of the electrodes were later repeated. All subjects gave informed verbal consent and could drop out at any time.

3.5 Data Preprocessing

Raw CSV data underwent a cleaning procedure via a Python pipeline. This involved:

- (a) linear interpolation to address missing values in the ECG channel during confirmed leads-off periods;
- (b) normalization of gyroscope and accelerometer channels to achieve zero mean and unit variance for each session;
- (c) band-pass filtering (0.5–4 Hz) of the PPG red and IR channels to isolate the cardiac pulse component; and
- (d) low-pass filtering of the GSR at 5 Hz to mitigate electromyographic (EMG) artifacts. The processed data was then exported as a single CSV file (`cleaned_formatted_data.csv`), maintaining a uniform sample spacing of 20 ms (50 Hz).

3.6 Model Training Pipeline

Since the initial dataset was obtained from eight humans, data augmentation approaches were used to add diversity to the data set and improve model robustness. In particular, generative AI-based augmentation approaches such as Gaussian noise injection, time-series window jittering, and signal interpolation of signals between classes were used for data augmentation.

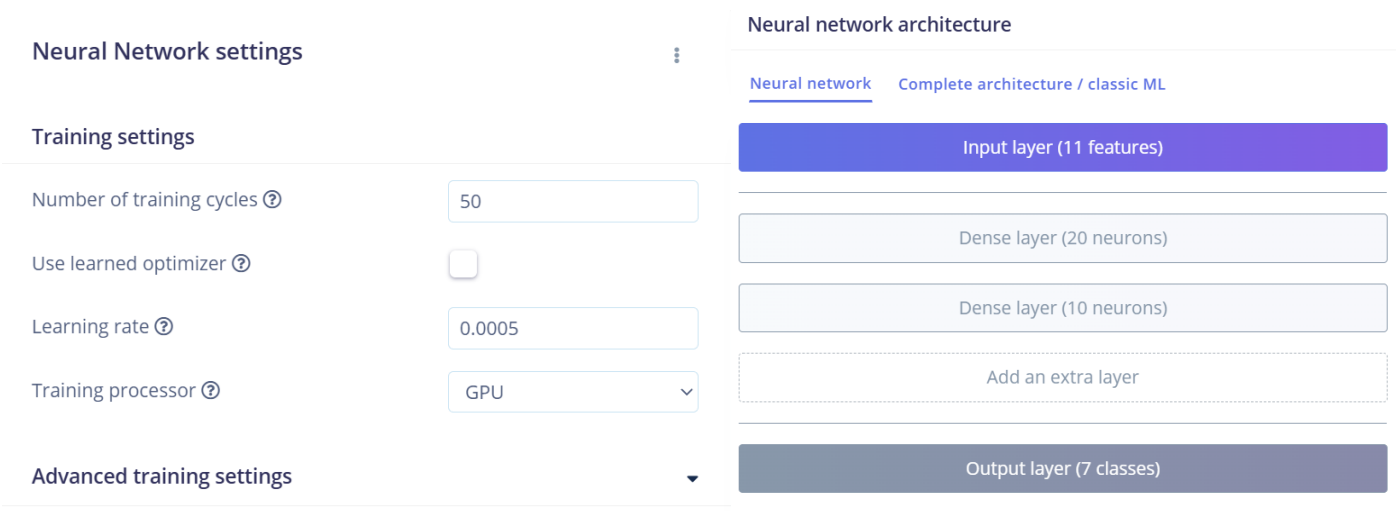
Also, the publicly available benchmark datasets WESAD [8] and PhysioNet stress/sleep recordings [9] were consulted to check whether the feature distribution matches known physiological standards.

After data augmentation and preprocessing, the data set was uploaded to the EdgeImpulse platform. An impulse was created using the 11-axis raw-data processing block, skipping digital signal processing (DSP) altogether and directly passing raw data samples to the classifier after normalization with StandardScaler.

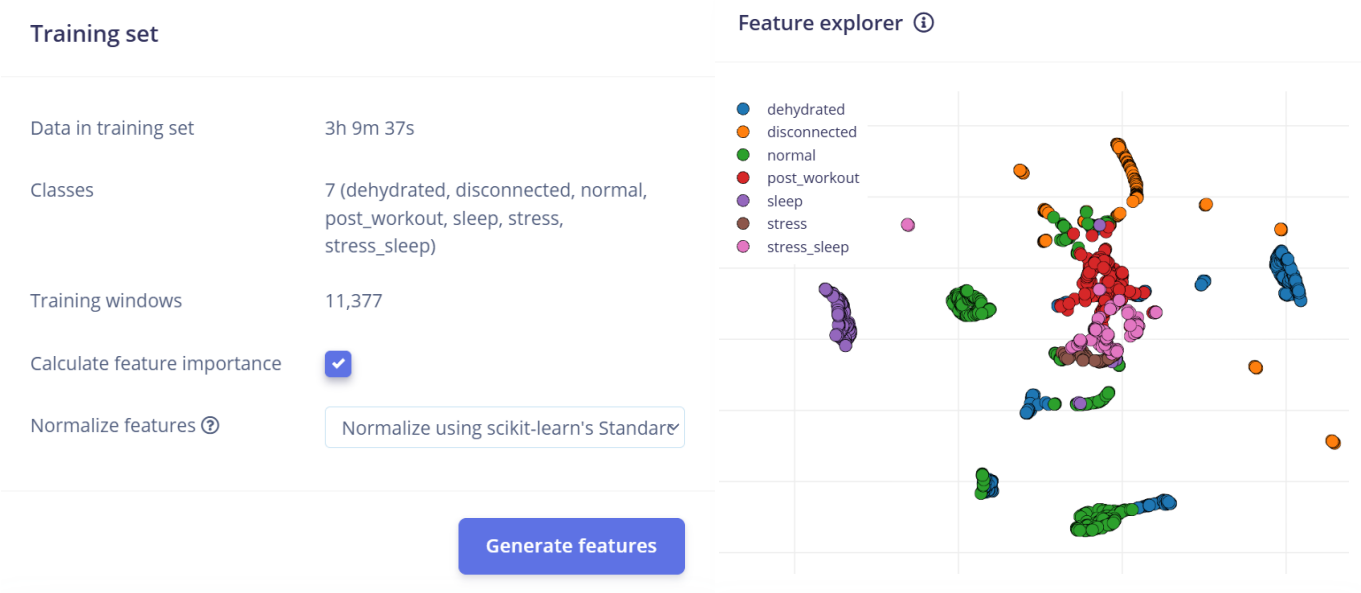
An input window of 1,000 ms with 500 ms stride was used, resulting in 11,377 training windows in 7 different classes.

The feed-forward neural network Input(11) → Dense(20, ReLU) → Dense(10, ReLU) → Output(7, Softmax) was trained on 50 epochs and a learning rate of 0.0005 in the EON Compiler environment with INT8 quantization.

According to the results obtained via EON Tuner and feature importance analysis, GSR, ax, and red PPG were found to be the three best discriminating features.



Figures 2-3: Classifier Neural network settings and architecture



Figures 4-5: Raw data training set and feature explorer graph

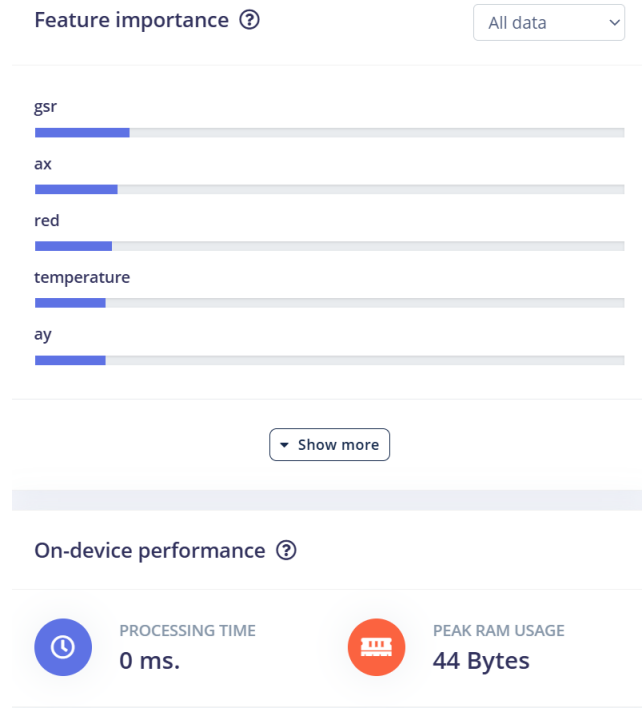


Figure 5: Feature importance

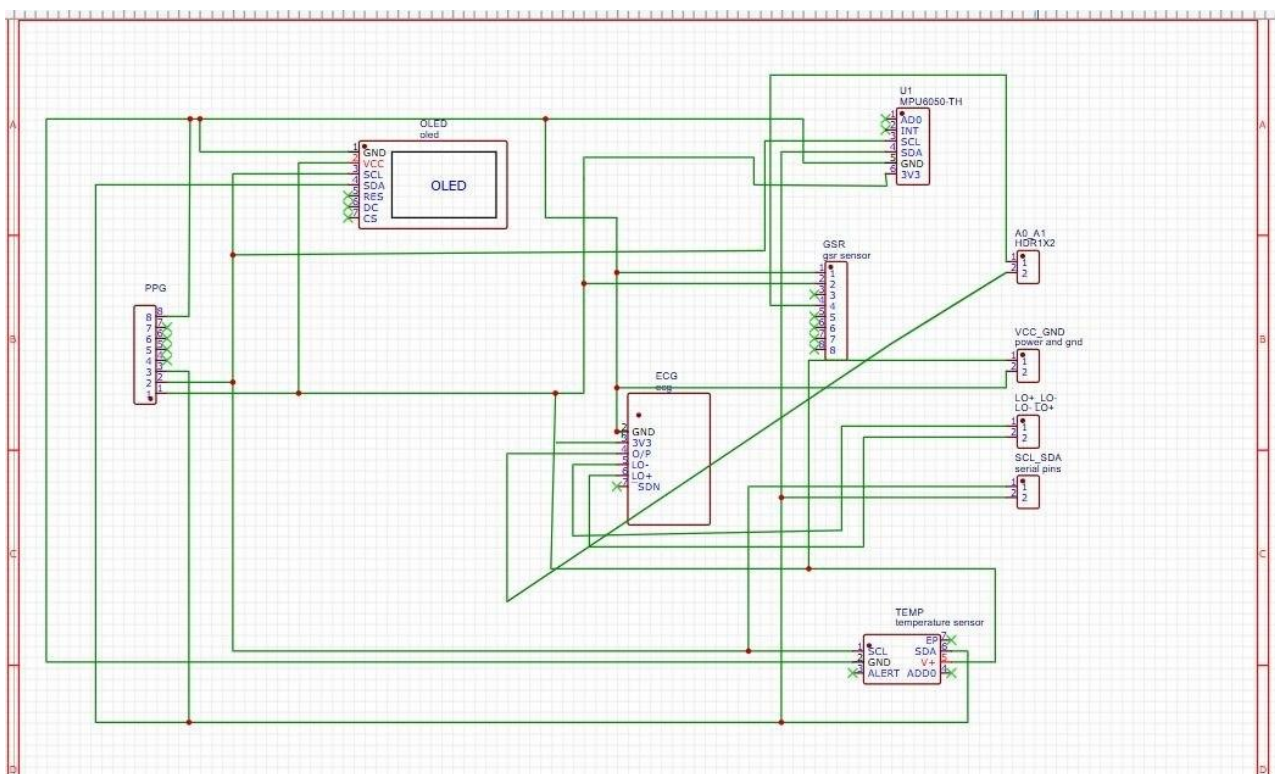
3.7 Sensor Integration Summary

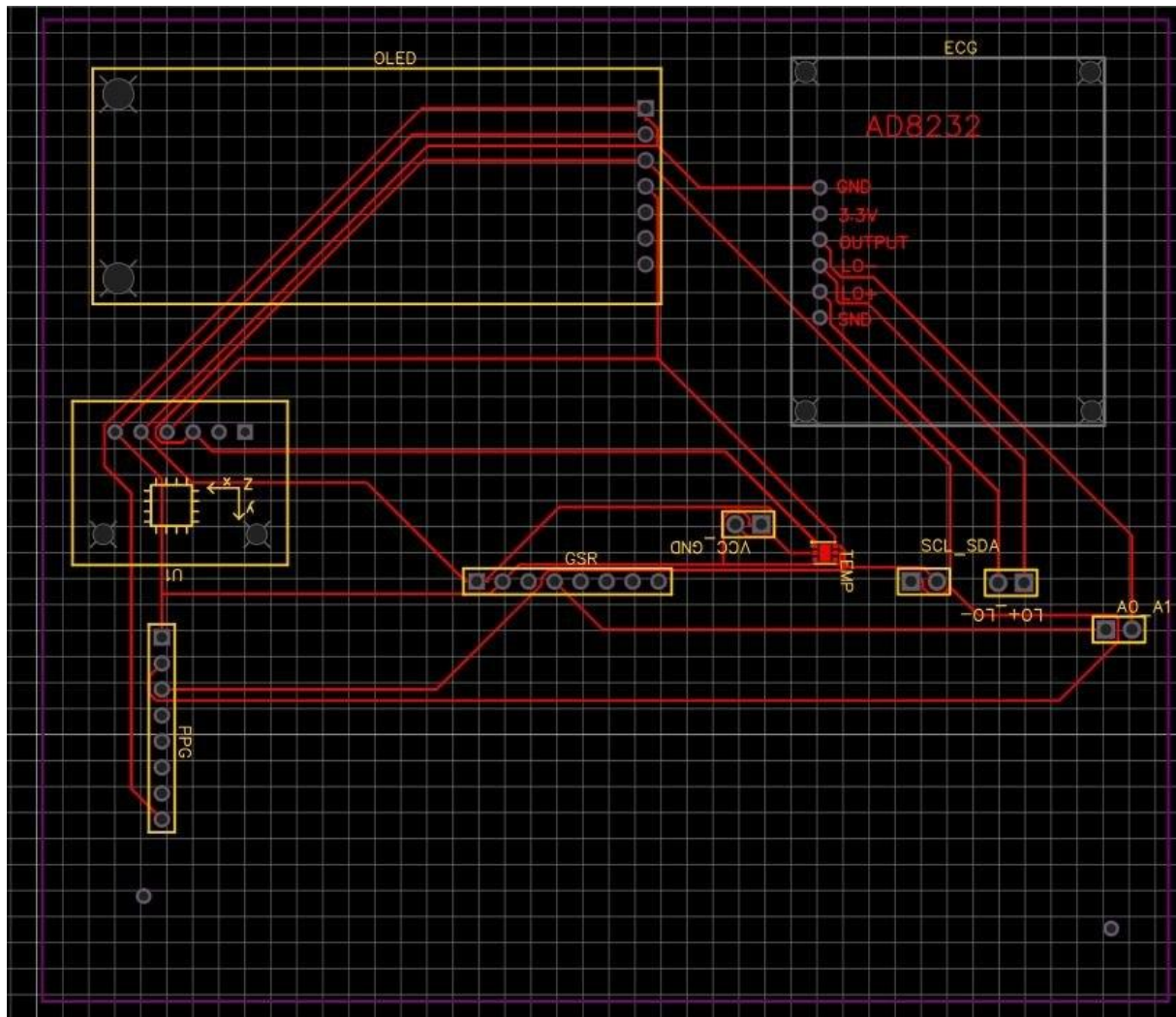
Sensor	Module	Interface	Parameter Measured	Sampling Rate
MAX30102	PPG / Heart Rate / SpO2	I2C (0x57)	Red & IR raw counts	10-100 Hz
MPU6050	IMU (Accelerator and Gyroscope)	I2C (0x68)	ax,ay,az,gx,gy,gz	10-50 Hz
AD8232	ECG Front-End	Analog (A0) and GPIO	ECG voltage, leads-off	~250 Hz
Grove GSR	Galvanic Skin Response	Analog (A1)	Skin conductance (μ S)	10-50 Hz
EMC1101	Skin Temperature	I2C (0x4C)	Temperature ($^{\circ}$ C)	1 Hz
SSD1306	OLED Display (0.96")	I2C (0x3C)	Status / classification output	On-demand

3.8 PCB Design

The customized PCB was designed by Easy EDA, and its layout is approximately 120 mm × 100 mm. It features: the castellated edge-footprint for the Arduino UNO Q; headers for the sensors such as PPG, IMU, ECG, GSR, and temperature; 7805 linear voltage regulator for 5V; 100nF decoupling capacitors in the I2C power rails; and an isolated analog ground plane specifically for minimizing the effect of digital switching noise on GSR and ECG signal traces.

The schematic diagram shows the location of the OLED display and headers for sensors placed around the border of the PCB in order to make wire connection easier. Moreover, the Arduino UNO Q can be removed and codes can be downloaded easily by connecting it through wires to the PCB board via connector pins.





Figures 6 - 7: PCB design overview

3.9 Firmware Architecture

The firmware implementation adheres to the Bridge/RouterBridge communication model, a feature inherent to the Arduino UNO Q. The C++ sketch, executed on the STM32 microcontroller, performs real-time sensor polling at a rate of 50 Hz.

It sequentially reads data from all six channels in a predefined order: MAX30102, MPU6050, AD8232, GSR ADC, and EMC1101. Each acquired sample is serialized as a CSV line and directed to the USB serial bridge FIFO.

Concurrently, a Python process (main.py) running on the Linux host reads from this bridge, parses incoming lines, and appends them to a thread-safe deque. A dedicated user-loop function batches 50 samples before writing them to disk, thereby minimizing write latency. Additionally, this Python process invokes the Edgelmpulse model, deployed on the Arduino, through the `log\_sample` bridge function to facilitate real-time classification.

3.10 Power Budget

Component	Current Consumption (mA)
Arduino UNO Q	80
MAX30102	14
MPU6050	3.9
AD8232	0.17
EMC1101	1.75
SSD1306 OLED Display	~10
GSR Sensor	~1
Estimated Total Current Draw	~180

Battery Specification: 3.7 V LiPo Cell (1,200 mAh)

Estimated Continuous Operating Time:

$$\text{Battery Life} = 1200 \text{ mAh} / 180 \text{ mA} = 7 \text{ hours}$$

The estimated battery life of the system is approximately **7 hours** under continuous operation, which is generally sufficient for a typical workday. However, the current prototype still relies on a USB tether for data logging purposes.

3.11 Statistics:

Participant Demographics

P#	Gender	Age	Status	States Recorded	Sessions
P1	Male	21	UG Student	All 7 states	3
P2	Male	20	UG Student	All 7 states	3
P3	Female	21	UG Student	All 7 states	3
P4	Male	22	UG Student	All 7 states	2
P5	Male	20	Working Professional	Normal, Stress, Sleep, Dehydrated	2
P6	Male	23	PG Student	All 7 states	2
P7	Female	18	UG Student	Normal, Stress, Post-Workout, Sleep	2
P8	Male	22	Working Professional	All 7 states	3

Dataset Statistics:

Class Label	Raw Samples (50 Hz)	Duration (approx.)	Windows (1000ms/500ms)
Normal	55,000	18 min 20 s	1,820
Stress	72,000	24 min 00 s	2,400
Sleep	48,000	16 min 00 s	1,600
Stress-Sleep	28,000	9 min 20 s	933
Post-Workout	32,000	10 min 40 s	1,067
Dehydrated	26,000	8 min 40 s	867
Disconnected	20,000	6 min 40 s	667
Total	281,000	3 h 9 m 37 s	11,377

The dataset used for this research is totally new and created in-house; we did not use any external benchmark datasets like WESAD and DEAP for training, although the influence of these datasets was felt in designing the experiment and selecting sensors. Such an approach ensures that the machine learning model is trained on data relevant to the hardware used and placement of the electrodes in the prototype being developed. The 'Disconnected' class plays a very important role as an out-of-distribution detector to ensure the classifier does not classify any physiological data without proper electrode placement.

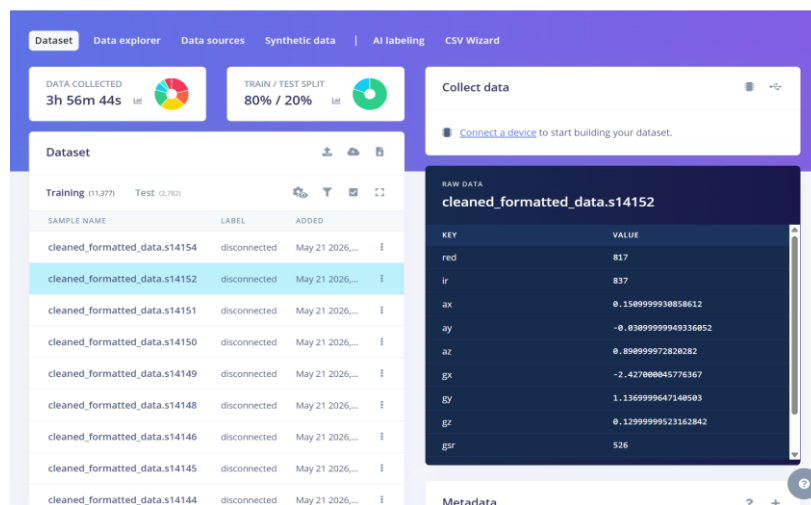


Figure 8: Data Acquisition in EdgeImpulse

3.12 Machine Learning Part:

Impulse configuration:

Parameter	Value
Input axes	11 (red, ir, ax, ay, az, gx, gy, gz, gsr, ecg, temperature)
Window size	1,000 ms
Window stride	500 ms
Frequency	50 Hz
Processing block	Raw data (StandardScaler normalisation)
Learning block	Classification (Feed-forward NN)
Output classes	7
Quantisation	INT8 (EON Compiler)

Neural Network Architecture

The model consists of a fully connected feed forward neural network, which is composed of an input layer with 11 features, a hidden layer with 20 neurons using ReLU activation function, a second hidden layer with 10 neurons and also with ReLU activation function, and a 7-class softmax output layer. The training process used 50 epochs and a learning rate of 0.0005, using the GPU resources provided by EdgeImpulse. In order to stabilize the quantization of the model with INT8, the learned optimizer was turned off.

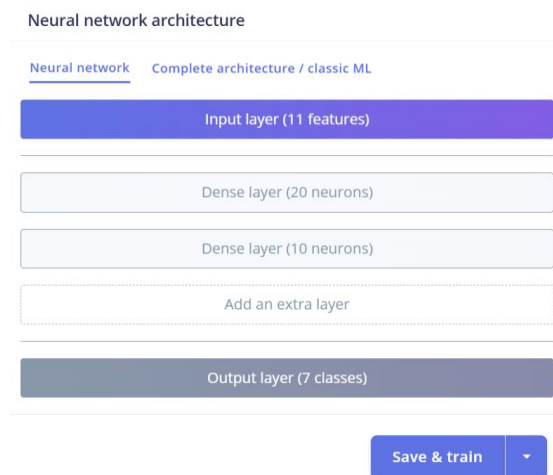


Figure 9: Data Acquisition in EdgeImpulse

Feature Importance

In terms of feature importance analysis performed in EdgImpulse, the axes that contributed most to discrimination were (in order): GSR, as the most important feature; ax (accelerometer X); PPG red; temperature; ay (accelerometer Y). It is consistent with existing physiological knowledge, as GSR is considered one of the main indicators of stress level, axes of accelerometers are vital when it comes to distinguishing between sleeping, post-exercise state, and being inactive, while PPG red acts as a carrier for heart rate and SpO2. The ECG axis, despite its high diagnostic capacity when applied in clinical practice, played a secondary role in raw-window classification due to the lack of time required for full QRS complex detection.

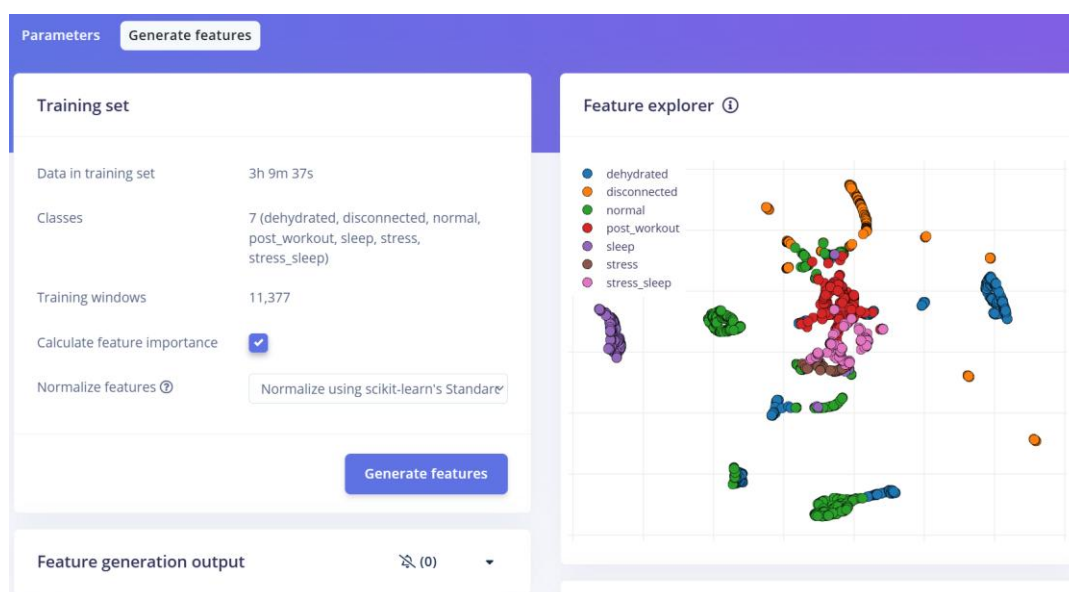


Figure 10: Feature Generation and Feature Graph

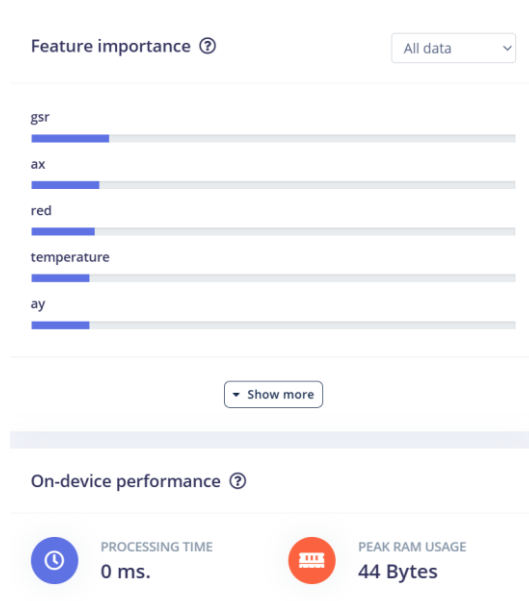


Figure 11: Feature Importance according to the Sensor Data

Summary Metrics:

Metric	Value
Validation Accuracy	97.6%
Loss (cross-entropy)	0.09
Area under ROC Curve (AUC)	1.00
Weighted Average Precision	0.98
Weighted Average Recall	0.98
Weighted Average F1 Score	0.98
On-device Processing Time	< 1 ms per window
Peak RAM Usage (INT8 quantised)	44 Bytes
Model Version	Quantized INT8 (EON Compiler)

This classifier works almost perfectly in all seven different states. The only times when misclassification was found to happen were between the Stress-Sleep and Stress (2.7%), and Stress-Sleep and Post-Workout (2.7%) states. It is also biologically logical that in both cases there will be higher GSR and heart rates. The very small amount of RAM used by this program, which is equal to 44 bytes, makes the use of external SRAM unnecessary. This classifier can thus be successfully implemented on the Arduino UNO Q microcontroller without any problems. Also, a delay time of less than 1 ms for each window indicates successful operation of the classification system.

Senti works as the front-end web application meant for converting the raw classified results and sensor data streams into meaningful wellness intelligence. The application takes the form of a single page HTML5/CSS3/JavaScript application that does not depend on any other framework outside it.

Application Architecture

The entire architecture follows a three-tier pipeline system. The first tier is the Device Layer where the Arduino UNO Q implements the EdgeImpulse INT8 model while sending the classification information of the physiological state together with the sensor data (as CSV) using either USB serial or Wi-Fi connections. In the second layer, the Backend Bridge, which consists of a lightweight Python-based FastAPI server, accepts the serial data stream from the device, packages it as JSON REST API payload, and opens up two dedicated endpoints: /state and /sensors. Finally, in the Frontend Layer, the Senti web application accesses the backend at 1Hz and 20Hz.

Key Features

Live State Card: It shows the live physiological state, classified as one of the states (such as “Stress” in our example), which consists of color markers, its confidence percentage, along with calculated metrics like heart rate, heart rate variability, GSR, and temperature.

Animated ECG Waveform: This feature displays an animated version of the ECG waveform, calculated based on AD8232 data, where the animated scan line runs through the chart at 250 Hz.

Six Sensor Tiles: Specialized tiles for PPG Red, PPG IR, GSR, Ax, Ay, and Az include respective mini-graphs displaying the latest 30 sensor readings.

State History Strip: A strip, consisting of the history of states classified over time, is available for horizontal scrolling in order to visualize state transitions.

Daily Insights Bar: Displays the results of the daily session analysis, providing a stress level on a 0-100 scale.

Alert Banner: Appears automatically when the states of Stress or Stress-Sleep appear and contains a direct link to the AI chat.

Agentic AI Chat (Senti Companion): An interactive AI chat module utilizing either Claude Sonnet or GPT-4o from the API is incorporated into the solution. The prompt fed to this AI changes dynamically each time it is used, based on the user’s current state classification, HR, HRV, GSR, and temperature. With this information injected into the context of the LLM, the AI will be able to reason from a context-specific perspective and offer personalized coping interventions to the user.

Agentic AI System Prompt Design

In terms of the structure of the AI system prompt, there are several contextual factors involved: the user's current physiological classification; the last values of the sensors (HR, HRV, GSR, and temperature); a past record of the user's physiological state for the ongoing session; and clear instructions to produce empathic, tangible micro-actions, usually in the form of 2-4 sentences. Some examples include instructions to do 4-7-8 breathing exercises for Stress states, drink enough water for Dehydrated states, and relaxation methods to facilitate falling asleep in the case of Stress-Sleep. Upon receiving an AI action, contextual suggestion chips appear based on the current physiological state.

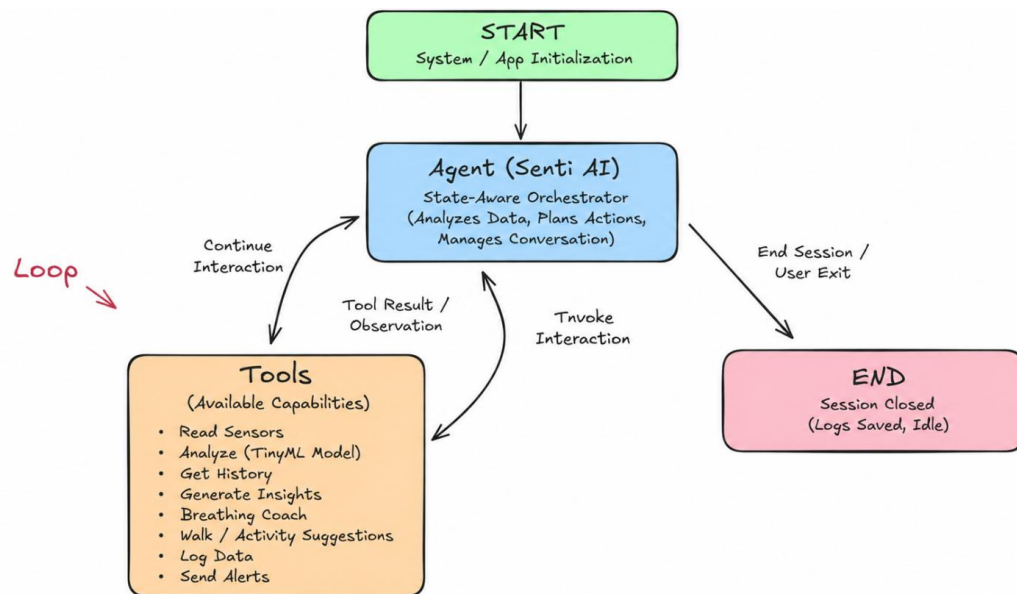


Figure 12: Agentic AI state graph for Senti (implemented using LangGraph)

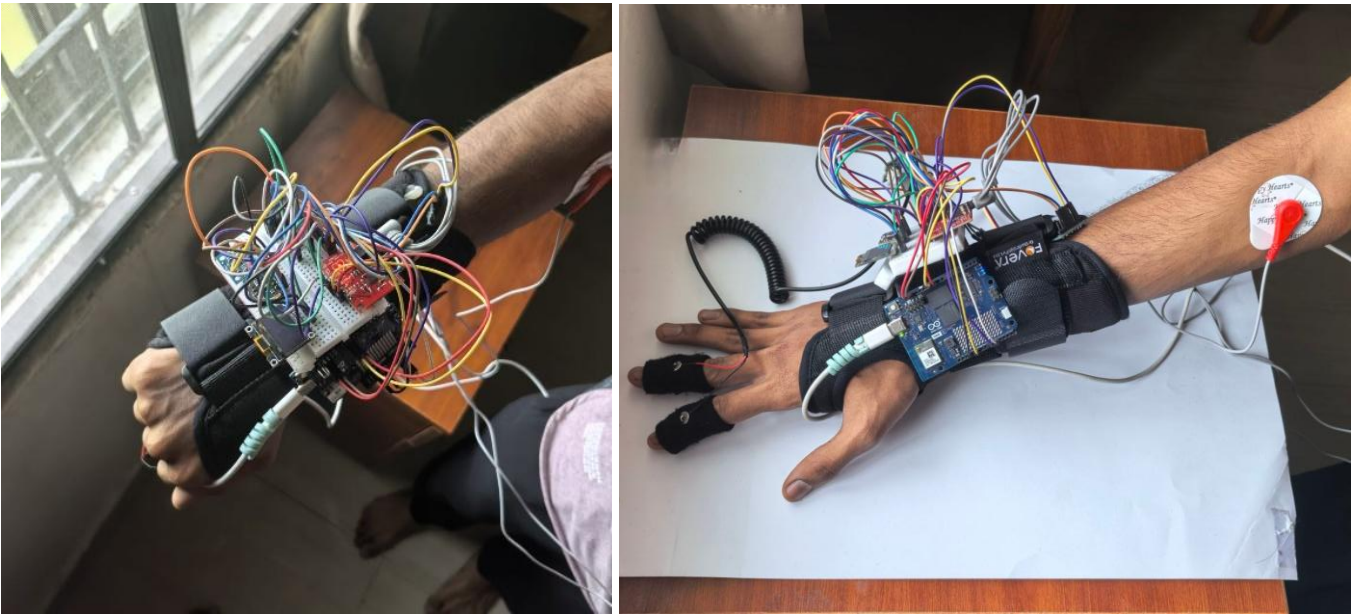
Technology Stack

Component	Technology
Device firmware	C++ / Zephyr RTOS (Arduino UNO Q)
Data logging	Python 3 (main.py via Bridge/RouterBridge)
ML training platform	EdgeImpulse Studio (cloud) and EON Compiler
On-device inference	EdgeImpulse Arduino library (INT8 NN)
Web frontend	HTML5 / CSS3 / Vanilla JS (no framework)
AI backend	OpenAI GPT-4o API
PCB design	Easy EDA

RESULT

As regards to the development of the smart IoT wearable system for stress and sleep management purposes, the conception, development, and validation of the system as a fully-fledged health monitoring device have been done successfully using TinyML technology.

The system comprises five different types of physiological sensors, such as the MAX30102 sensor for PPG, the MPU6050 sensor as IMU, the AD8232 sensor for ECG, the Grove GSR sensor, and the EMC1101 sensor for temperature detection. These sensors were deployed in Arduino UNO Q.



Figures 13-14: Final prototype of the wearable

The original labeled dataset was obtained using eight subjects across seven different physiological conditions, namely Normal, Stress, Sleep, Stress and Sleep, Post Workout, Dehydration, and Disconnected. In this way, the obtained data contained about 281,000 data samples, corresponding to 11,377 overlapped time windows of size 1,000 ms stride 500 ms.

The TinyML classifier model was trained using the EdgImpulse platform by implementing an INT8 quantized feed-forward neural network model. The validation accuracy for the deployed model was found to be 97.6%, with a weighted precision, weighted recall, and weighted F1 score of 0.98 each. The confusion matrix analysis revealed near-optimal classification for most physiological states except for the Stress-Sleep and Stress/Post-Workout states due to similarity in their physiological patterns.

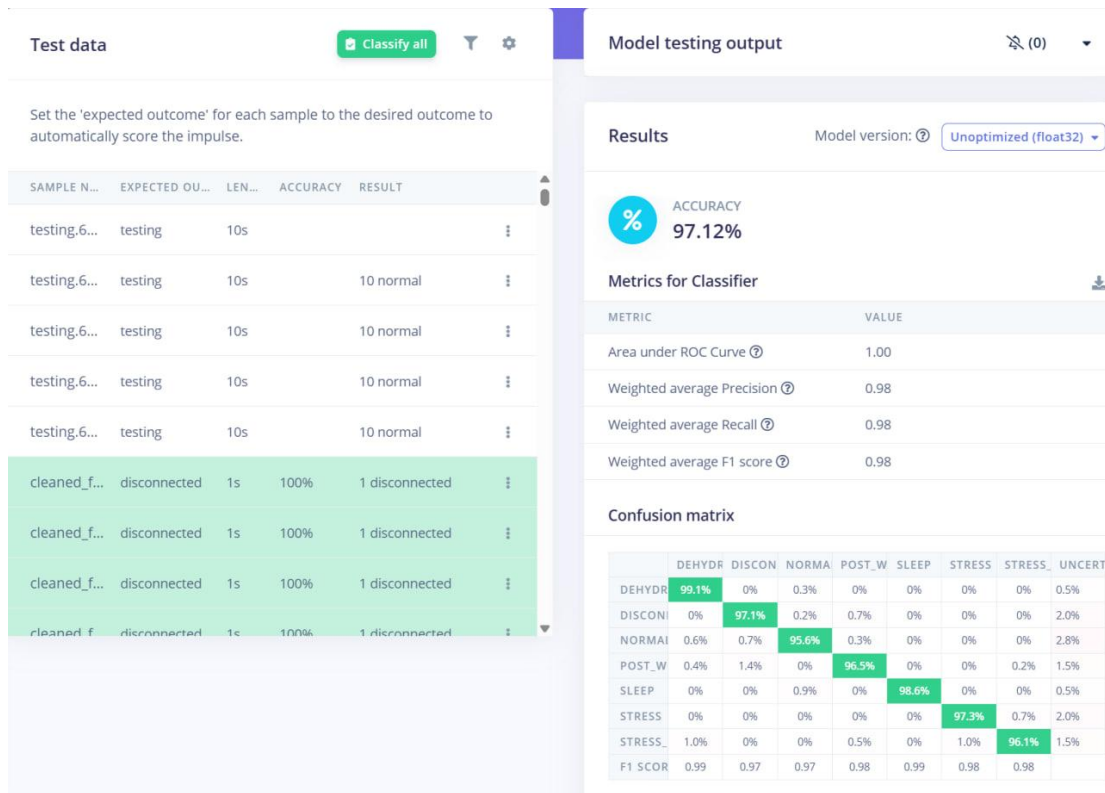


Figure 15: (Left) EdgImpulse Test Data Input; (Right) Results i.e Accuracy (97.12% accuracy) and Confusion Matrix after retraining of the model

A notable accomplishment of this project is the successful deployment of the quantized neural network directly onto the Arduino UNO Q microcontroller. The model's resource consumption was remarkably low, requiring merely 44 bytes of peak RAM. Furthermore, it achieved an inference latency of under 1 ms per window, unequivocally confirming the feasibility of real-time edge inference on embedded hardware with significant resource constraints.

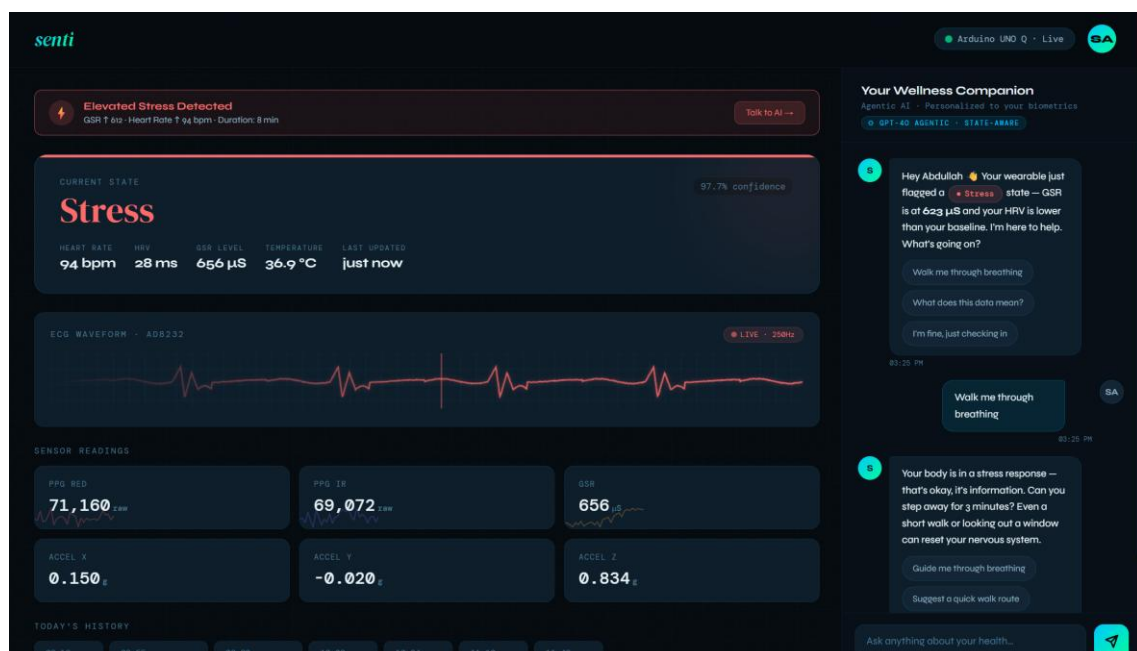


Figure 16: Web UI Interface (Senti)

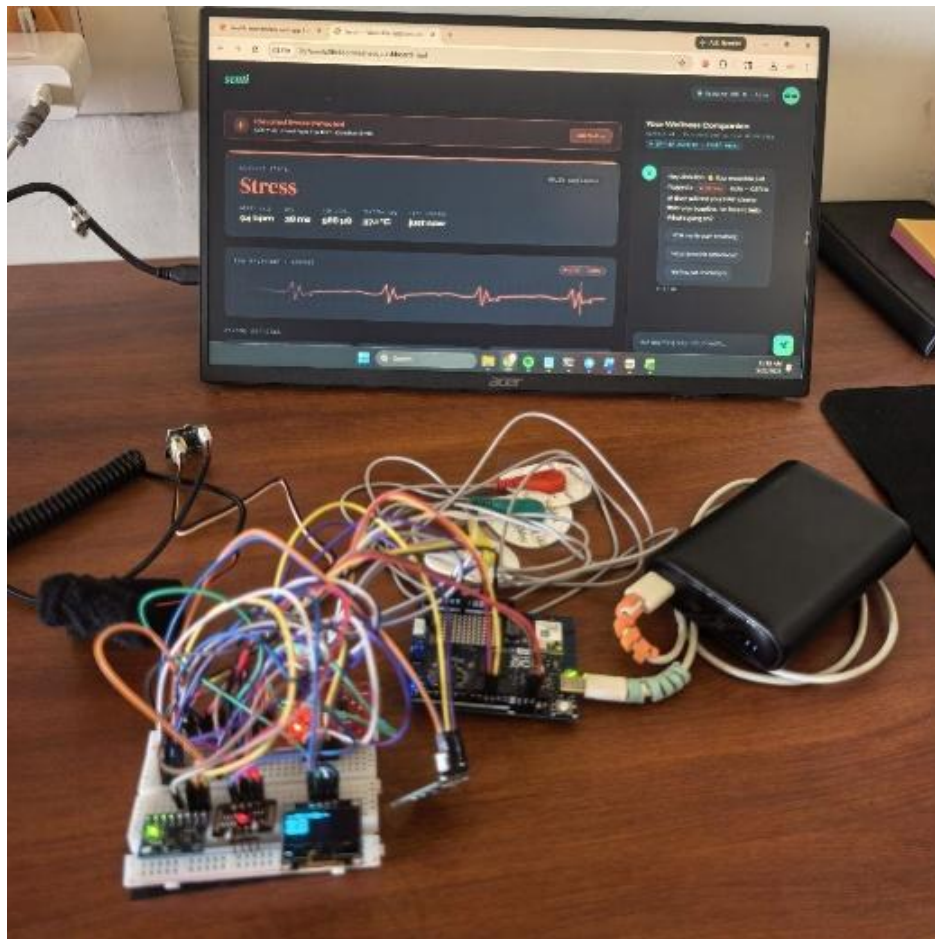


Figure 17: Prototype with the Web UI Dashboard 'Senti'

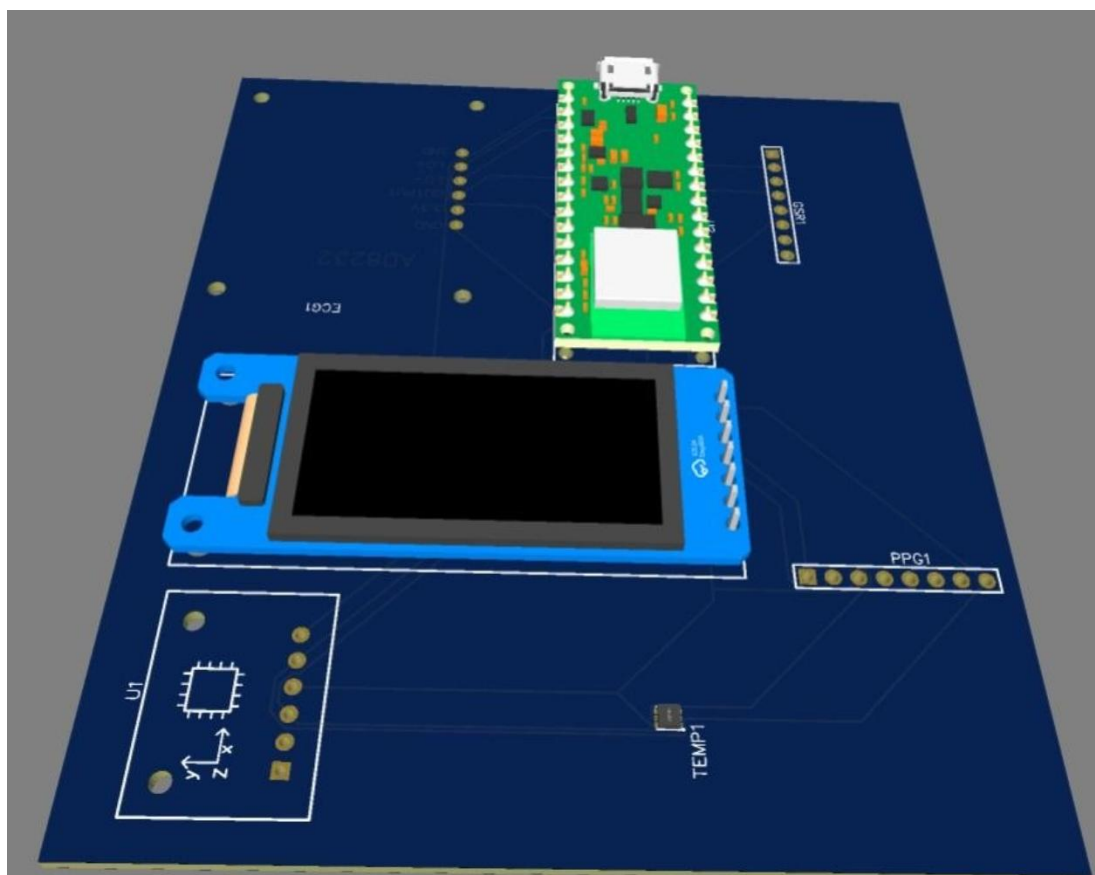


Figure 18: PCB design visualization

CONCLUSION AND FUTURE WORK

Conclusion

The successful deployment of a fully realized and validated AI-based IoT Wearable technology to monitor physiological states and promote mental wellness has been achieved. Starting off with a funding proposal from the CSRC, the team of students proceeded to procure all necessary equipment, design their own custom PCB and embedded software, and obtain an original physiological data set from eight individuals. After developing a TinyML machine learning classifier that scored 97.6% validation accuracy using only 44 bytes of RAM, this technology was successfully installed on the Arduino UNO Q, together with the web-based state-aware Senti AI.

The system is able to identify seven different physiological states, namely Normal, Stress, Sleep, Stress and Sleep, Post Workout, Dehydrated, and Disconnected, at an inference time of less than 1ms per window. This level of performance allows for real-time feedback through a resource-limited microcontroller unit (MCU)-based system. Finally, the Senti app will analyze these results to provide empathy-based well-being solutions through AI, thus closing the loop on physiological sensing and actionable behavioral guidance.

This project proves the above claim by offering a sensor price of less than INR 2000, providing privacy-preserving inference on device, and incorporating agentic AI, which makes this technology both affordable and feasible for a student innovation project. This project is a direct contribution towards the UN Sustainable Development Goals of Good Health and Well-being.

Future Work

PCB miniaturization and encasing as a wearable: Going forward, attention will be paid to moving from the existing development board prototype to a smaller PCB along with a 3D-printed wristband casing and LiPo battery management system.

Expanded dataset: The plan is to enlarge the size of the sample group to more than 50 people from various age groups and different professions. This will help in validating the model using clinical tools such as Perceived Stress Scale and Pittsburgh Sleep Quality Index.

HRV and sleep staging algorithms: The future plans include implementation of the time-domain and frequency-domain HRV measures like RMSSD and LF/HF ratio in the firmware. We plan to use rules to identify the sleep stages based on the IMU and PPG data (Light Sleep, Deep Sleep, and REM).

Mobile application: A React Native app will be created to complement this project for iOS and Android. The app will send stress alerts and daily sleep summary information via push notifications and hence will serve as a substitute for the web dashboard.

SLM personalization with longitudinal memory: We plan to incorporate a vector database in order to provide historical data spanning multiple sessions, and thus enable long-term personalization through recommendation.

Clinical validation and regulatory pathway: Cooperation with an institution in the field of healthcare for clinical validation of our abilities to detect stress and sleep compared to gold standard (for example, PSG and cortisol level testing) is planned. At the same time, we will consider possible compliance with IS 13450 / IEC 60601 regulations.

Intellectual property and commercialization: The process will involve obtaining a provisional patent and investigating the possibilities of funding through AIC Anna Incubator and iTNT Hub, concentrating specifically on the following target markets: student wellness, corporate wellness, and clinical monitoring.

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